



Machine Learning–Based Behavioral Analysis Reveals Persistent Sex-Specific Outcomes in a Chronic Stress Mouse Model

Poster #D8



Kaitlyn Huynh¹, Mohan Yang², Carlos Rodriguez-Ortiz¹, Steven Grieco¹, Krisha Thakker¹, Zhaoxia Yu², Xiangmin Xu¹

¹Department of Anatomy & Neurobiology, School of Medicine, University of California, Irvine, Irvine, CA, USA

²Department of Statistics, Donald Bren School of Information and Computer Sciences, University of California, Irvine, Irvine, CA, USA

Abstract

- Chronic stress can produce lasting behavioral changes, but how these adaptations evolve over time and differ between sexes can be further investigated.
- Traditional behavioral assays rely on predefined endpoints and human observation, potentially missing subtle behavioral changes and introducing observer bias.
- MoSeq uses unsupervised machine learning to identify sub-second behavioral “syllables,” enabling detailed characterization of spontaneous behavior over time.
- Examining male and female mice with MoSeq may reveal sex-specific patterns of behavioral adaptation to chronic stress.

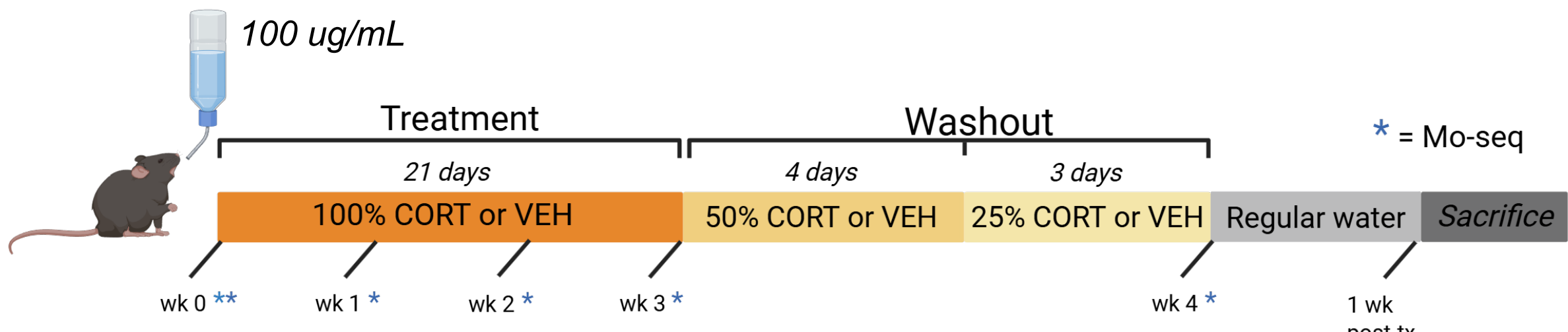
Experimental Design

Study Cohorts

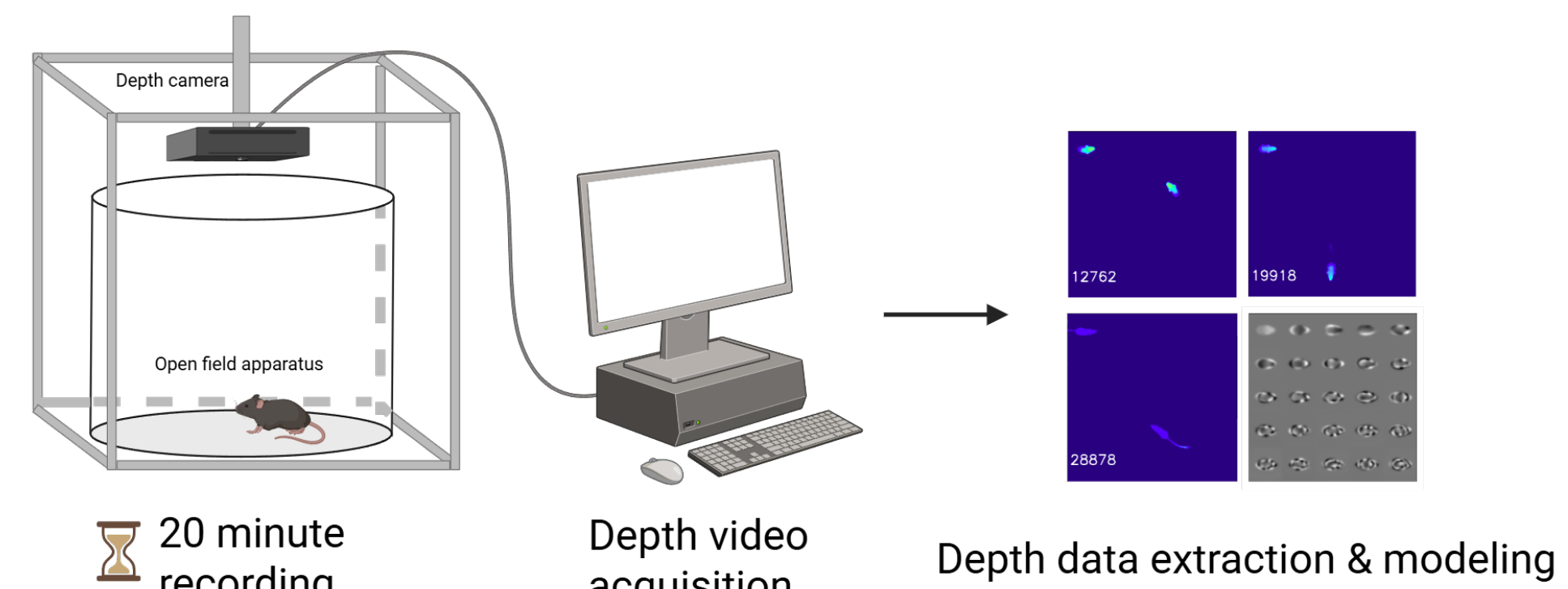
Sex	VEH	CORT
♂ Male	n = 23	n = 20
♀ Female	n = 12	n = 12

Age: 3 months

Experimental Timeline



Motion Sequencing Workflow



Why MoSeq?

- Unsupervised behavioral analysis
- Sub-second behavioral dynamics
- Longitudinal tracking of spontaneous behavior
- Detects subtle behavioral organization

CORT alters whole-animal movement and behavioral profiles.

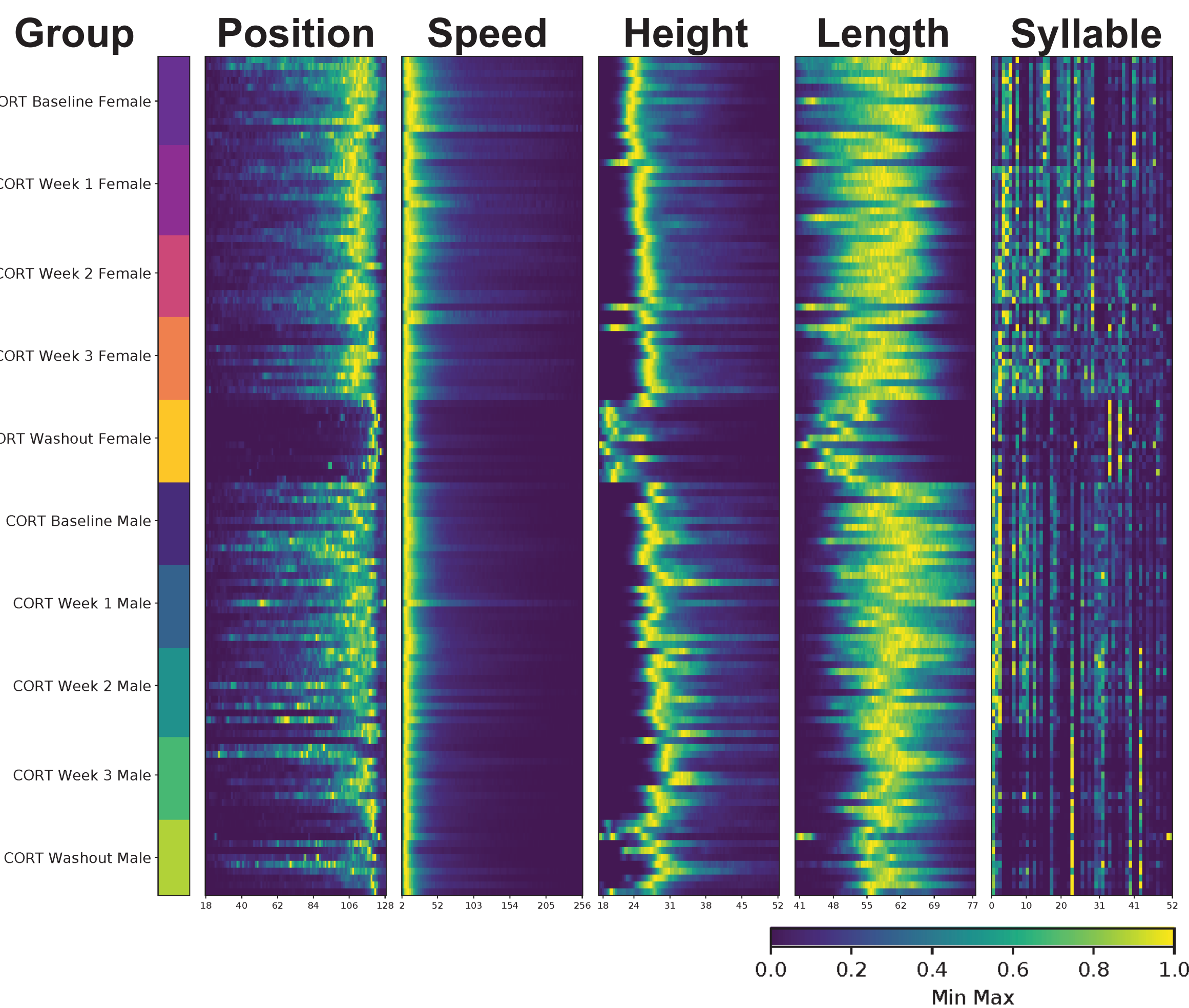


Figure 1. Behavioral fingerprints of kinematic features extracted from MoSeq analysis, including distance from center, speed, length, height, and syllable usage probability.

Unsupervised MoSeq analysis identifies recognizable mouse behaviors.

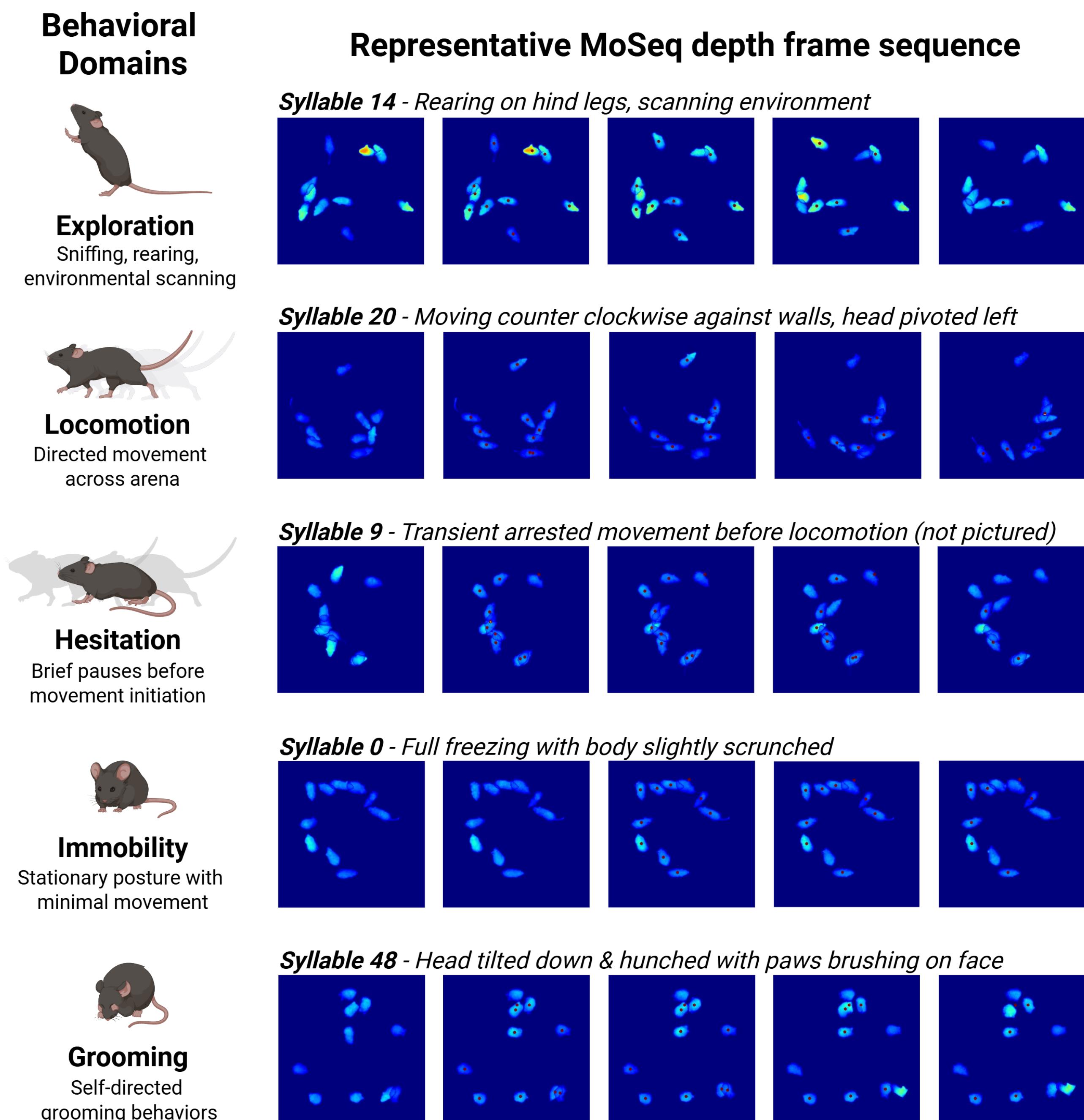


Figure 2. Representative MoSeq syllables frame sequences that correspond to five behavioral domains. Each sequence is a composite of 10 mice from unique recording sessions, selected based on similarity to the average kinematic features of the syllable. Depth coloring shows body position relative to the sensor (warmer = proximal, cooler = distal).

Results

Sex-specific behavioral profiles emerge following CORT exposure.

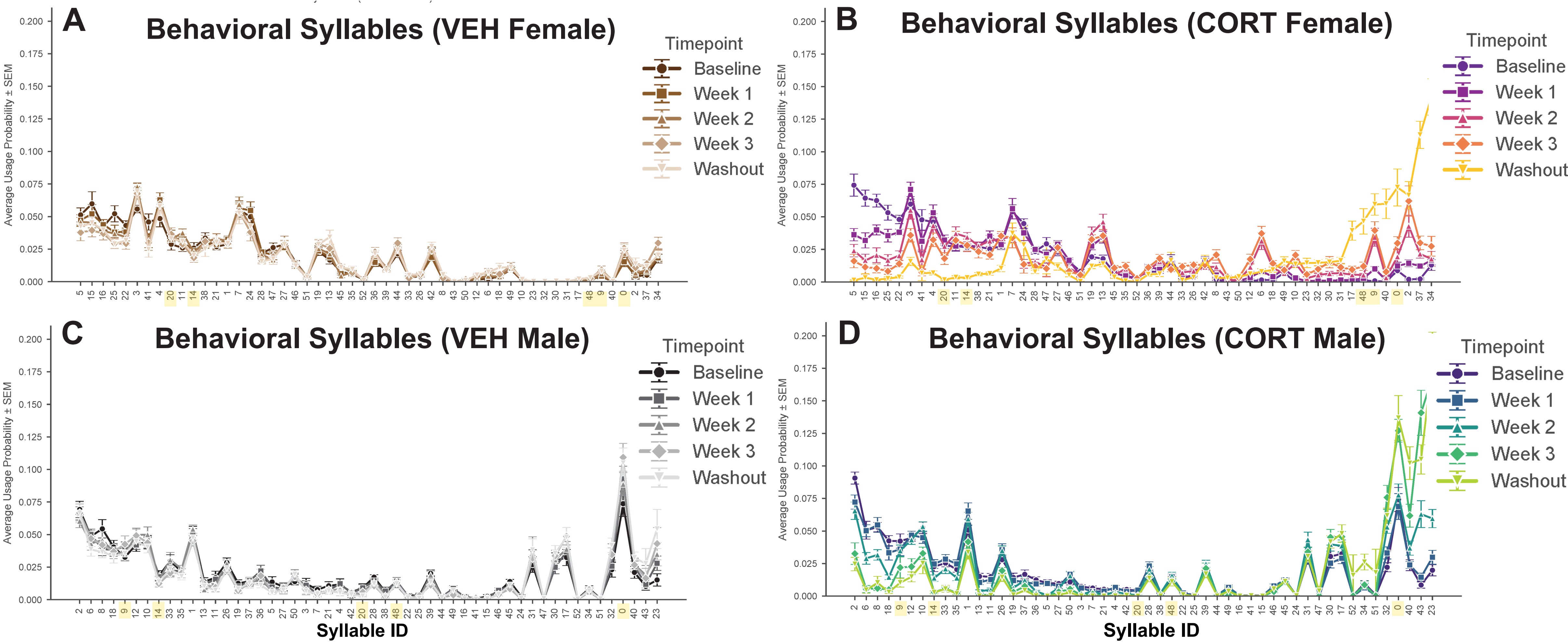
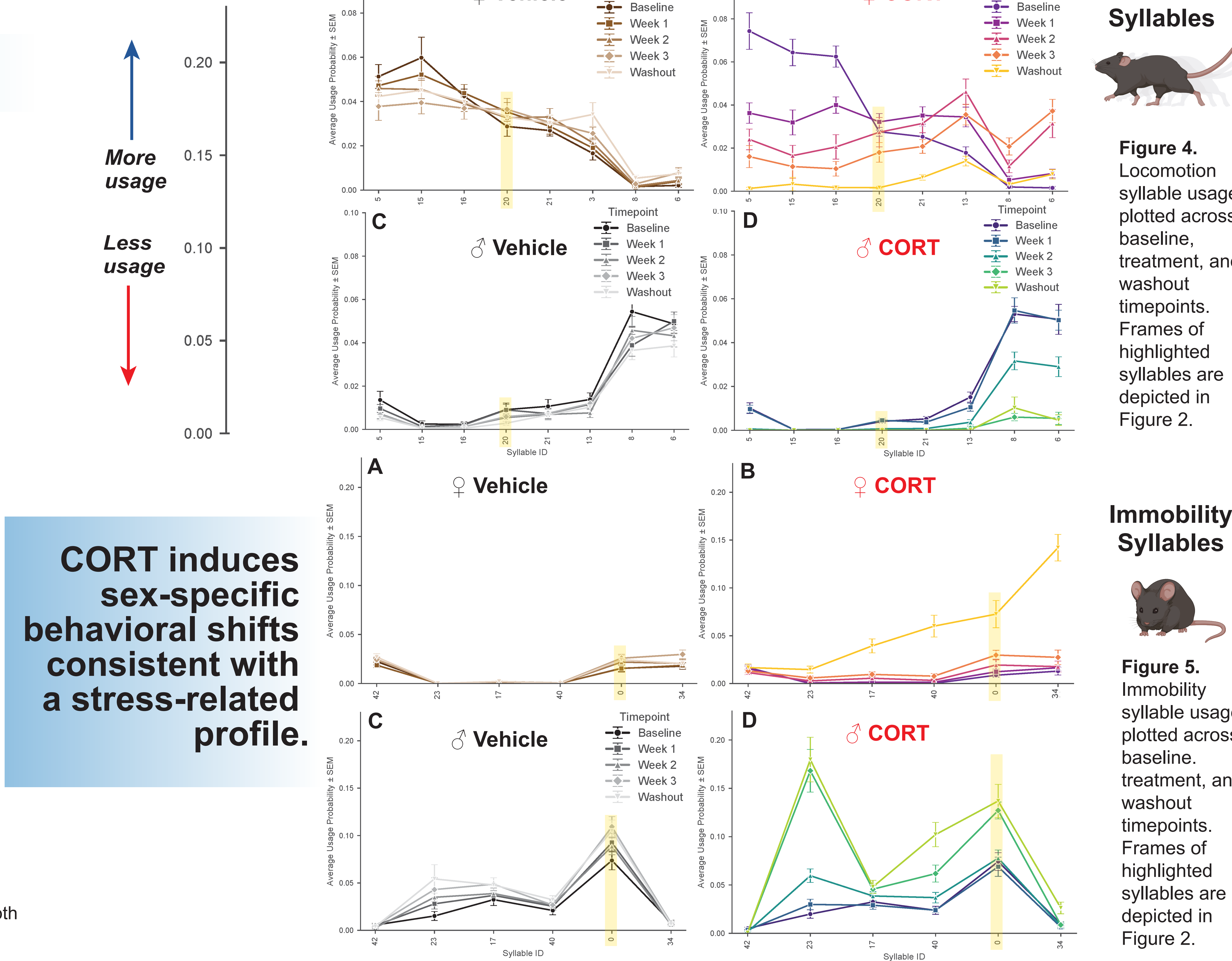


Figure 3. Average behavioral syllable usage probability across baseline, CORT treatment timepoints, and washout. The MoSeq model identified 50 behavioral syllables, each assigned a numerical ID (1–50). Panels show female vehicle (A), female CORT (B), male vehicle (C), and male CORT (D).



Locomotion Syllables

Figure 4. Locomotion syllable usage plotted across baseline, treatment, and washout timepoints. Frames of highlighted syllables are depicted in Figure 2.

Immobility Syllables

Figure 5. Immobility syllable usage plotted across baseline, treatment, and washout timepoints. Frames of highlighted syllables are depicted in Figure 2.

Conclusions

- MoSeq can reveal **fine-scale changes** in spontaneous behavior induced by CORT treatment.
- CORT produced sex-specific changes across behavioral syllables, domains, and treatment timepoints.
- Sex-specific behavior alterations *persisted* following CORT treatment cessation.
- Machine learning–based approaches show promise for capturing subtle behavioral adaptations that may be overlooked by conventional behavioral measures.

Acknowledgements

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Figures were made utilizing Biorender.

References

- ¹ Lin et al., 2024 — MoSeq methodology and analysis framework
- ² Wiltshcko et al., 2015 — Original MoSeq approach and behavioral syllable discovery